NATHANYL L. FREESE

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EDUCATION

Embry-Riddle Aeronautical University (ERAU) – Daytona Beach Campus **Exp. Grad:** December 2026
Bachelor of Science, Aerospace Engineering **GPA:** 3.97/4.00
Tau Beta Pi Engineering Honor Society

WORK EXPERIENCE

Collins Aerospace, Melbourne, FL May 2026 – Present
Systems Engineering Intern

- Developed Python automation to execute 301 transponder test files, reducing test runtime by 50%
- Evaluated an internal AI test case generation tool across 80+ requirements, providing feedback to improve accuracy
- Built a data dictionary supporting RAG-based test case generation by defining requirement inputs and ranges
- Supported requirements verification and test procedure development using Jama, Confluence, and GitLab
- Updated legacy transponder test procedures and verified test equipment to support sustainment and system transition

KBR & LZ Technology, Houston, TX January 2025 – August 2025
Engineering Intern

- Developed Python and PyQt5 tools for antenna tracking, blockage analysis, and Orion/Starliner telemetry visualization
- Optimized the Gateway visualization tool, achieving a 13× overall performance boost and 21× faster graphics rendering
- Led development of the Starliner visualization tool with integrated telemetry overlays for operational simulations

Thermodynamics Tutor, ERAU, FL January 2024 – January 2025

- Tutored over 400 students seeking help in Thermodynamics and Heat Transfer each semester
- Created and provided targeted problem sets, exam reviews, and custom lesson plans to boost students' understanding

PROJECT EXPERIENCE

Two-Stage Methalox Crewed Lunar Mission, ERAU, FL January 2026 – Present
Project & Systems Lead

- Led a 6-person engineering team in the design of a two-stage LOX/CH4 launch vehicle for a crewed lunar mission
- Directed requirements, subsystem integration, and engineering trade studies through Preliminary and Critical Design Reviews
- Conducted propulsion, CFD, thermal, and structural analyses to evaluate engine performance and verify design requirements
- Developed detailed launch vehicle CAD assemblies in CATIA V.5 and Onshape, supporting subsystem integration

Heavy Lift UAV Project, Daytona Beach, FL April 2026 – October 2026
CAD & Simulation Engineer

- Designed frame components in Fusion 360 and SolidWorks, optimizing strength-to-weight ratio and manufacturability
- Contributed to the design of a 300+ lb payload attachment system and motor mount, developing CAD models for integration
- Developed an Unreal Engine 5 simulation integrated with PX4 to support autonomous flight testing and validation

Embry-Riddle Future Space Explorers and Development Artemini Rocket, ERAU, FL August 2024 – January 2025
Manufacturing Member

- Modeled a scaled-down Artemis rocket in CATIA V.5 and SolidWorks, using OpenRocket to optimize flight performance
- Collaborated with a 15-member team on design modifications and structural integration

Experimental Rocket Propulsion Lab Spectre Rocket, ERAU, FL August 2022 – May 2024
Hardware Member

- Contributed to the design, assembly, and testing of an active stabilization system, including canard integration, for a high-powered rocket within a 10-member engineering team
- Modeled structural components in Fusion 360, soldered avionics hardware, and analyzed test data in MATLAB

SKILLS

CAD & Simulation: CATIA V.5, SolidWorks, Fusion 360, Creo Parametric, Onshape, NASTRAN, ANSYS Fluent, Unreal Engine 5
Programming: Python, MATLAB, C, PyQt5, Linux CLI, GitLab, GitHub, VS Code
Engineering Tools: NASA CEA, PX4, Jama, Jira, Confluence, Agile/Scrum, Microsoft Office